Chest X-ray Generation with a VAE

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Code: `github.com/CiciAa777/285hw1-vae-implementation`

1 Model and Hyperparameter Settings

An unconditional convolutional VAE [1] is trained on chest X-rays [2]. The encoder is a CNN that maps an image x to the mean $\mu_\phi(x)$ and log-variance $\log \sigma^2_\phi(x)$ of a diagonal Gaussian posterior $q_\phi(z \mid x)$. Latents are sampled with the reparameterization trick, $z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon$, $\epsilon \sim \mathcal{N}(0, I)$, and a transposed-CNN decoder maps z to the reconstruction $\hat{x}$. The training objective is to minimize the β-weighted negative ELBO,

$$\mathcal{L} = \|x - \hat{x}\|_2^2 + \beta \, D_{\mathrm{KL}}\big(q_\phi(z \mid x) \,\|\, \mathcal{N}(0, I)\big), \tag{1}$$

where the first term is the summed pixel-wise squared error and β trades reconstruction against closeness to the prior. To generate images, we sample $z \sim \mathcal{N}(0, I)$ and decode it. Table 1 lists the data, training setup, and the hyperparameters tested.

Table 1: Data, training settings, and tested hyperparameters.

Data	5,216 train / 16 validation / 624 test; grayscale, bilinear resize, values in $[0, 1]$
Architecture	D stride-2 blocks, channels $16, 32, 64, 128$ (for D=4); mirrored decoder
Training	Adam, lr 10^{-3}, batch 32; 30 epochs (64^2) or 40 (128^2); best-validation checkpoint
Tested values	$H \in \{64, 128\}$, $d_z \in \{32, 64\}$, $\beta \in \{0.5, 1, 2\}$, $D \in \{3, 4, 5\}$
Seeds	42 for all runs; 43 for the repeated β comparison

2 Training Loss Curves

Figure 1 shows the total, reconstruction, and KL losses for the three-block model ($H = 128$, $d_z = 32$, $\beta = 0.5$). Training losses decrease steadily; validation curves are noisy because the validation set has only 16 images.

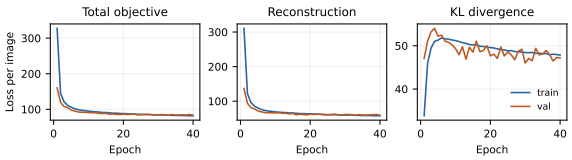


Figure 1: Loss curvces over 40 epochs ($D = 3$, $H = 128$, $d_z = 32$, $\beta = 0.5$, seed 42).

3 FID/IS Scores and Hyperparameter Results

Following the Inception-based workflow of [3], we generate 624 images from the prior (generation seed 123) and compare them with the 624 test images. The Inception Score (IS) [4] uses generated images only, and the Fréchet distance [5] compares generated and real statistics. Our Fréchet values use the 1,000-d softmax outputs rather than the standard 2,048-d pooled features, so we denote them FD_{1000} rather than FID. Validation MSE (per-pixel mean) is also reported as a held-out reconstruction measure.

We tuned one factor at a time, in three stages, fixing each choice before moving on: (1) image resolution and latent dimension, (2) KL weight β with a second seed, and (3) network depth. The baseline was $H = 128$, $D = 4$, $d_z = 32$, $\beta = 1$ (Table 2).

Table 2: All completed runs, grouped by stage. s: training seed. Lower MSE and FD_{1000} and higher IS are better; best values in bold.

Stage	Run	H	D	d_z	β	s	Val. MSE	FD_{1000}	IS
	Reference	128	4	32	1	42	.004090	.2462	1.8396
1	Lower resolution	64	4	32	1	42	.006006	.2351	1.4971
1	More latent dims	128	4	64	1	42	.004163	.2459	1.8723
2	$\beta = 0.5$	128	4	32	.5	42	**.003562**	.2440	1.9334
2	$\beta = 2$	128	4	32	2	42	.004997	.2518	1.7648
2	Reference, repeat	128	4	32	1	43	.004195	.2766	1.9183
2	$\beta = 0.5$, repeat	128	4	32	.5	43	.003586	.2334	1.9352
3	Shallower	128	3	32	.5	42	.003628	**.2150**	1.9546
3	Deeper	128	5	32	.5	42	.003598	.2785	**1.9675**

Stage 1: resolution and latent dimension. Going from 64^2 to 128^2 lowers MSE by 32% and raises IS from 1.50 to 1.84, while FD_{1000} is slightly worse. The comparison is confounded, since the two resolutions differ in model capacity and loss scale. Doubling d_z to 64 changes all metrics only slightly. *Takeaway:* keep $H = 128$ and $d_z = 32$.

Stage 2: KL weight. Lowering β from 1 to 0.5 reduces MSE by 13% (seed 42) and 15% (seed 43) and improves both generation metrics in each seed, while $\beta = 2$ is worse on all three metrics in its one run. The FD_{1000} of the $\beta = 1$ reference moves from .2462 to .2766 between seeds, so differences in FD_{1000} below roughly 0.03 are within seed noise. *Takeaway:* $\beta = 0.5$ is the most consistent improvement.

Stage 3: network depth. Validation MSE is nearly flat across depths 3, 4, and 5 (.00363, .00356, .00360). Depth 3 has the lowest FD_{1000} (.2150) and depth 5 the highest IS (1.9675) but the worst FD_{1000} (.2785); the spread is on the order of the seed noise above. Parameter counts also differ (1.672M, 1.140M, 1.791M for $D = 3, 4, 5$), so we cannot attribute differences to depth alone. *Takeaway:* depth has no clear effect in this range.

4 Generated Chest X-ray Examples

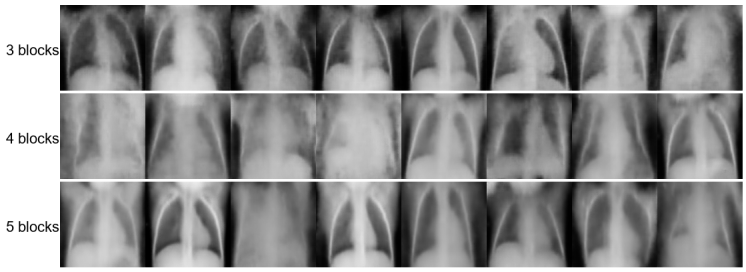


Figure 2: Prior samples for $D = 3, 4, 5$ (rows); $H = 128$, $d_z = 32$, $\beta = 0.5$, seed 42, generation seed 123.

References

[1] D. P. Kingma and M. Welling. Auto-Encoding Variational Bayes. ICLR, 2014.

[2] P. Mooney. Chest X-Ray Images (Pneumonia), Kaggle.

[3] PyTorch-Ignite. GAN evaluation using FID and IS.

[4] T. Salimans et al. Improved Techniques for Training GANs. NeurIPS, 2016.

[5] M. Heusel et al. GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium. NeurIPS, 2017.